AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science and Engineering

Faculty of Science and Technology

CSC4118: COMPUTER GRAPHICS

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Section: [H], Group No: 06

Project Report On

Animating the Spirit of Hatirjheel

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Table of Contents

Introduction:

This project illustrates a 2D animated view of Hatirjheel using OpenGL and GLUT. The scene captures the essence of Hatirjheel with its wide lake, bridges, modern buildings, and greenery. Dynamic elements such as moving cars and buses on the roads, boats gliding over the water, clouds floating across the sky make the environment feel lively. Trees sway gently to mimic natural motion. User interactions are enabled through both keyboard and mouse, allowing changes in animation speed, day–evening-night transition, and control over different visual effects. The simulation relies on basic geometric shapes, transformation functions, and timers to achieve smooth movement and transitions. With a clean modular design, each object of the scene is drawn and animated through separate functions. This project provides an engaging visualization of Hatirjheel while reinforcing essential principles of computer graphics.

Keyboard controls:

D- Day mode

E- Evening mode

N- Night mode

H- Increase moving vehicle's speed

L- Decrease moving vehicle's speed

P- Pause and resume animation

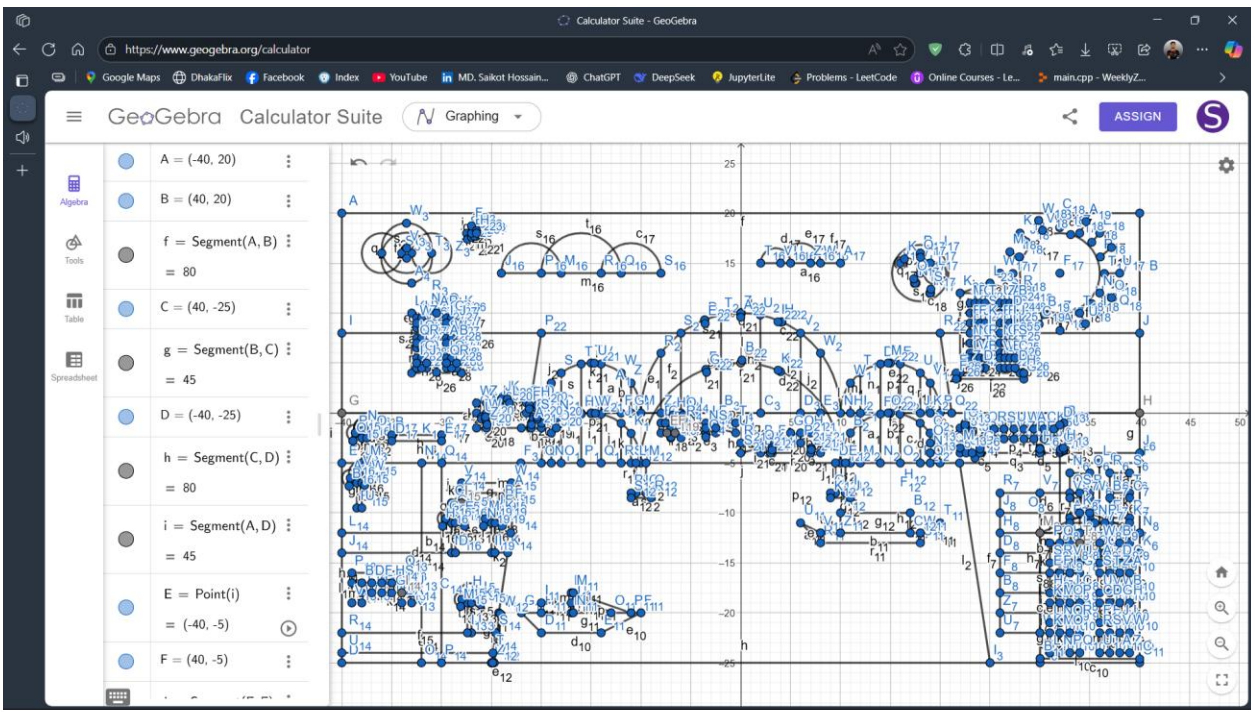
Mouse control:

Right click- Start raining

Left click- Stop raining

This simulation was constructed with C++, OpenGL, and GLUT, which combined offer the ideal resources for producing 2D visuals and fluid real-time animations. I've used important computer graphics techniques in this project, such as object transformations, matrix operations for element placement, and timer routines that make everything come to life. Keyboard and mouse commands are used for the interactive elements, and scene polishing is achieved using visual effects like transparency blending. This project demonstrates the artistic process of creating an appealing animated environment as well as the technical aspects of graphics programming.

Project graph:



List of Objects:

SL#	Object ID	Object Name
01	Obj_01	Rain drops
02	Obj_02	Birds
03	Obj_03	Cloud1
04	Obj_04	Cloud2
05	Obj_05	Moon
06	Obj_06	Sun
07	Obj_07	Sky
08	Obj_08	Bridge Structure
09	Obj_09	Bridge Lights
10	Obj_10	Car1
11	Obj_11	Car2
12	Obj_12	Car3
13	Obj_13	Bus
14	Obj_14	Road
15	Obj_15	Boat
16	Obj_16	Speed boat
17	Obj_17	Trees
18	Obj_18	Building1
19	Obj_19	Building2
20	Obj_20	Building3
21	Obj_21	Lake water
22	Obj_22	Park seat
23	Obj_23	Park road

List of Functions:

SL#	Object Name	Function Name
01	Rain drops	drawRaindrops_D()
02	Birds	bird_D()
03	Cloud1	Cloud_D_1()
04	Cloud2	Cloud_D_2()
05	Moon	Moon_D()
06	Sun	Sun_D()
07	Sky	Sky_D()
08	Bridge Structure	Bridge_Structure_D_Func()
09	Bridge Lights	BridgeLight_D()
10	Car1	Car_D()
11	Car2	Car_D()
12	Car3	Car2_D()
13	Bus	Bus_D()
14	Road	Bridge_D()
15	Boat	Boat_D_1()
16	Speed boat	Boat_D_2()
17	Trees	Tree_D()
18	Building1	Building1_D()
19	Building2	Building2_D()
20	Building3	Building3_D()
21	Lake water	Lake_D()
22	Park seat	Park_D()
23	Park road	Road_Park_D()

List of Animation Functions:

SL#	Animation Function ID	Animation Function	Object/Scene
01	Animation_01	Rain_Update_D()	Rain fall
02	Animation_02	Bird_Update_D()	Flying birds
03	Animation_03	Cloud_Update_D()	Cloud Movement
04	Animation_04	Boat_Update_D()	Boat Movement
05	Animation _05	Vehicle_Update_D()	Cars,Bus
06	Animation_06	keyboard_D()	Day, Evening, Night, Pause, Resume, Vehicle speed up/down

Contribution:

Member Name	Implemented Functions	Implemented Animations	Percentage of Contributions
Md. Saikot Hossain	All drawing functions	All animation functions	100%

Conclusion:

This project successfully demonstrates how computer graphics concepts can be applied to create an interactive and visually appealing simulation of Hatirjheel. By combining geometric modeling, transformations, and animation techniques, the scene captures both the natural and urban aspects of the area in a simplified yet engaging form. The integration of user interaction enhances the experience, allowing viewers to explore different times of day and dynamic environmental effects. Overall, the project not only highlights the beauty of Hatirjheel but also serves as a practical exercise in understanding and implementing fundamental principles of 2D animation and OpenGL programming.